

Evaluation of automated table extraction for historical infectious-disease surveillance data

Eden Ofek Nalian¹ Steven C. Walker¹ David J. D. Earn¹

October 3, 2026

¹Department of Mathematics & Statistics, McMaster University, Hamilton, Ontario, Canada

Corresponding author: eden.nalian@gmail.com (Eden Ofek Nalian)

Abstract

Most historical epidemiological data remain undigitized. This paper evaluates automated table extraction as an alternative to manual data entry, assessing *Amazon Textract* on a large dataset of historical infectious disease records: extraction accuracy, error characteristics, and efficiency relative to manual transcription. Cell-level agreement was evaluated on the 1956 tables (53,872 evaluated cells): content-only agreement between the automated and manually transcribed data was 88.5% overall and 92.9% among cells numeric in the manual reference transcription (a secondary figure of 95.6% obtains if the denominator is instead restricted to cells numeric on both sides). Downstream epidemiological comparisons (disease peaks, growth rates, cross-correlations) span all three years of the dataset, 1956–1958. Most residual cell mismatches had a Levenshtein distance of one or two; larger discrepancies were concentrated in rows affected by disease-name or row-alignment failures, and differences in the downstream epidemiological summaries were generally modest. Extraction runtime was short relative to an extrapolated manual-entry estimate, but total person-time, including post-processing, was not measured; the post-processing time we did record was

a small proportion of the total time previously required for the whole task. This paper illustrates the potential and limitations of a human-in-the-loop workflow for digitizing historical tabular data of this kind.

1 Introduction

Weekly notifiable-disease reports have been collected in printed form for well over a century, but most remain undigitized and unavailable for computational analysis. Records that have been converted to machine-readable form have typically relied on manual transcription – accurate, but slow enough to constrain which archives are digitized at all, and not itself error-free (section 5 documents an instance where the manual transcription used in this study contained an error that the automated extraction did not). Automated table extraction is an obvious alternative, but for a researcher fitting models to the resulting series, the relevant question is not how well a tool reads cells in the abstract: it is whether the epidemiological quantities computed from the extracted data match those from a trusted transcription, and, where they do not, whether the discrepancies can be detected without one.

Software that converts images of tables into machine-readable data, such as comma-separated value (CSV) files, typically decomposes the problem into distinct tasks, though different authors characterize them differently [1]. We organize our work around three tasks:

Table detection: Locating tables in documents.

Table structure recognition: Parsing the row and column layout of tables.

Content recognition: Identifying the content of table cells and headers.

Table detection and layout are surprisingly difficult for computers given how easy they are for humans [2], a difficulty rooted in decades of OCR research [3] and still evident in widely used open-source engines such as Tesseract [4]. Modern document-AI systems – including large vision-language models applied directly to spreadsheets and tables – continue to report

substantial failure modes on structurally irregular inputs even on contemporary, cleanly scanned material [5, 6], a limitation that compounds on historical documents, where period typography, physical degradation, and non-standard layouts are the norm. Software that produces usable results – data similar enough to the scanned document to support similar downstream conclusions – must succeed at all three tasks reasonably well; a single under-performing sub-task is often enough to compromise the result, as we discuss in section 5.

Historical health-surveillance archives are a natural and under-served target for this technology. Project Tycho is the most developed prior effort of this kind, digitizing over a century of United States notifiable-disease reports [7, 8]; Canada’s own notifiable-disease surveillance system has a comparably long paper record that, until recently, remained largely undigitized [9, 10]. Largely absent from this literature, and from table-extraction benchmarks generally, is evaluation against an independent transcription with downstream, analysis-level epidemiological quantities – not only cell-level accuracy – as the endpoint: benchmark accuracy on curated corpora does not say whether the peak timing, growth rate, or cross-correlation an epidemiologist computes from an extracted series would match a trusted transcription, or whether the discrepancies could be caught without one. Reference transcriptions are also conventionally treated as an infallible standard against which extraction is scored; as we show in section 5, they can themselves contain errors that a careful comparison can surface.

Using 156 weekly tables from the Canadian Disease Incidence Dataset (CANDID; 1956–1958) [11], we ask: how closely does a documented, staged Textract-based extraction and post-processing workflow reproduce an independently produced manual transcription at the cell level (evaluated here for the 1956 tables); which classes of error survive each processing stage; and how well are standard epidemiological summaries – peak timing, peak size, initial growth rate, and between-disease cross-correlation – preserved before and after targeted human review, for the disease pairs we examine? A useful table-extraction tool must perform all three tasks above while balancing speed, accuracy, and cost against manual entry: extraction time plus correction time should be smaller than the time to do the whole task

manually, and no more expensive. Even a tool that scored perfectly on these criteria would not obviate a researcher’s own reading of the results [12] – verifying a workflow like this one still requires a baseline of human time and attention.

Amazon Textract, the extraction service used throughout this study, is a cloud-based machine learning service that recognizes table, form, and key-value structure in scanned documents, in contrast to OCR tools that read only raw text; it is accessed either through a browser interface or programmatically via an API (section 2 gives implementation details).

2 Data extraction and Post-Processing

This section summarizes the process of extracting data from a scanned table to a CSV file, and the post-processing needed to achieve usable results. The basic extraction procedure does not depend on the specific dataset and applies to any PDF repository of files (dataset details are in section 3.1); further details on Textract are in Amazon’s official documentation [13]. For running AWS Textract and the provided extraction script, see section 6.4.

2.1 API Call to Textract

AWS Textract can be accessed programmatically via API calls with any popular high-level programming language; we use Python, via AWS’s official SDK, `boto3`. A researcher calls Textract’s `AnalyzeDocument` endpoint and receives a flat, nested JSON structure of nested `BLOCK` objects (tables, cells, and words, each with a confidence score) that must be reassembled into the row/column structure a human would recognize; the full object structure is given in Supplementary Material (S3). Once reassembled, this process can be applied to multiple tables at once by iterating through a repository of scanned table files and returning the outputs in separate CSV files.

2.2 Post-Processing

Though it is trivial for humans to distinguish cell values from headers, Textract relies on an OCR engine and does not differentiate between the two: headers are not explicitly labelled, merged cells rely on `RowSpan/ColumnSpan`, and column meaning must be inferred by position. Post-processing is therefore essential for meaningful analysis.

Our post-processing normalized the cell data to match the original PDF tables' structure and cell types: trimming and removing whitespace, and converting cell values into strings.

Recognizing header values versus cell contents depends on the specifics of each dataset; we used the proportion of cell values in a row that were strings, and, through manual inspection of the original PDF dataset, labelled rows with more than 50% string content as headers. This proved successful: headers were removed correctly in almost every extracted table, with one exception caught and corrected manually.

Another major issue was rows shifted by one cell value, which invalidates an entire row's comparison against the manual data, especially for numerical values. This is fairly easy to detect once comparing cells, so a step was added to check row length against the row above it; this improved results when matching against the manual data (further described in section 4).

Cells were also corrected to known values based on the original dataset via dataset-specific coding, described in section 5.3.

3 Dataset Criteria and Selection

This section describes the dataset used to assess the quality of Textract-extracted data. Beyond the general tool-selection criteria in section 1 (accuracy, time, and cost relative to manual entry), the dataset had to satisfy one further requirement: it had to already have a manual transcription, so automated extraction could be checked directly against human data entry.

3.1 Canadian Disease Incidence Dataset

The Canadian Disease Incidence Dataset (CANDID) is a summary of notifiable-disease cases in Canada, by province and territory, maintained by the Dominion Bureau of Statistics (the predecessor of Statistics Canada) [11]. The data was stored in a fixed, typed template, so most cell contents are easily legible to humans.

We chose this dataset chiefly because it had already been independently transcribed by hand [11], enabling the accuracy measures in section 4; as detailed there, we treat that manual transcription as a reference for comparison, not a verified gold standard. We also chose it for the qualities of the table itself, described below.

We extracted data from the tables covering disease reports from 1956 to 1958, three years in total. There is one table per week, with columns for each province and territory recognized at the time plus country totals; each province’s columns are further split into cases reported for the current week, the previous week, and the median of the previous 5 years.

Each row corresponds to a specific disease, 28 rows in total. Most rows correspond to a single disease (e.g. the first row, Chickenpox), but some diseases span multiple rows for additional data – e.g. row 17 splits Tuberculosis into Pulmonary and Non-Pulmonary. See Week 1 of CANDID in fig. 1.

The table in fig. 1 is easily interpretable by a human observer, but poses several problems for an OCR engine such as Textract. Common problems in the cell structure and content, typical of many historical data documents, include (and lead to the problematic results in section 5):

- Merged cells: many rows contain merged cells (e.g. the Canada case totals, the province names, and the cumulative-total header), making row lengths inconsistent across the table.
- Header values for the current week, previous week, and median are printed vertically, while the rest of the table is printed horizontally.

Summary of Cases of Notifiable Diseases in Canada, as reported by the Provincial Health Departments for the week ended January 7, 1956, in comparison with the previous week¹, and the 5-year

(1) Figures adjusted for delayed reports. (2) Exclusive of Yukon and the Northwest Territories. (3) Total accumulated to the end of the current week from the first week ending in January of the year stated. (4) Includes cases where type was not specified. (5) Scarlet fever includes cases of streptococcal sore throat. (6) Type not specified. (7) Includes chancroid, granuloma inguinale and lymphogranuloma venereum. .. Not reportable.

THE SITUATION IN THE UNITED STATES (Source: "MORBIDITY AND MORTALITY WEEKLY REPORT", U.S. National Office of Vital Statistics; Number of reported cases for the current week (previous week in parentheses):

Diphtheria,	39	(54)	Measles	4,443	(3,725)	Poliomyelitis, acute	120	(119)	Poliomyelitis, without paralysis,	XX	(XX)	Smallpox,	-	(5)	Undulant fever,	14	(14)
Encephalitis infections,	20	(22)	Measingspoccal infections,	58	(77)	Poliomyelitis, with paralysis,	XX	(XX)	Scarlet fever (5)	XX	(XX)	Typhoid fever,	20	(23)	Whooping cough,	XX	(XX)

Only 39 cases of diphtheria were reported this week as compared with a high of 104 for the week ended December 10, 1955. Almost half (17) of the cases were reported in the West South Central States. These and 2 Southern States, Alabama (8 cases) and Florida (4), accounted for all but 10 of the total cases.

EPIDEMIC REPORTS: Measles - Nfld. reports epidemic at Harbour Main.

9005-501
18-1-56

Figure 1: Week 1 table of the Canadian Disease Incidence Dataset. Source: Dominion Bureau of Statistics, 1956; reproduced under Crown copyright's 50-year term (expired at the end of 2006 for a 1956 publication), placing this material in the public domain in Canada.

- Missing weekly data is marked “.”; weeks with no cases are marked “-”.
- Footnote markers: cells may carry a parenthetical number (1–6) referring to a footnote on data collection or provenance. For example, “5(5)” for P.E.I. under report week, row “Scarlet Fever”, means 5 cases, with footnote (5) reading “Scarlet Fever includes cases of streptococcal sore throat.”
- “The situation in the United States,” printed at the bottom of each week’s table, is not in tabular format and so was not processed by Textract (a correct table-detection outcome, section 1) – but is valuable information the tool did not capture.

4 Statistical Measures

4.1 Testing Accuracy of Textract Data Compared to Manually Entered Data

To assess the accuracy of the CANDID data digitized by Textract, we compared it with the corresponding data digitized by humans in McMaster’s Theoretical Biology group, using several metrics.

We refer to this human-digitized dataset as the *reference transcription* rather than a verified gold standard. It is an independently produced transcription, not a ground truth: like any manual entry process, it is itself subject to error, and we do not assume by definition that it is correct. Where possible, disagreements between the two sources are adjudicated against the original source scan rather than resolved in the reference transcription’s favour (see section 5.3 for an example in which this adjudication instead identified an error in the reference transcription).

4.1.1 Simple Metrics

The most straightforward metric, the *exact matched-cell proportion*, is the percentage of cells from the Textract dataset exactly matching the reference transcription – not a measure of cells correctly digitized, but of cells whose content *and* location (row and column index) are both equivalent. We applied this metric to the raw extracted data, then again as the data became progressively more processed through the methods in section 2.2.

Inclusion rule. We define the evaluated cell set reproducibly, following the implementation in `postprocessing/combine.ipynb`: each cleaned weekly CSV pair (`data/textract_cleaned/1956`, `data/manual_cleaned/1956`) is compared position-by-position over the overlapping row and column range, i.e. `min(n_rows)` by `min(n_cols)` between the two files for that week; any rows or columns beyond that overlap (e.g. from a length mismatch between the two sources) fall outside this range and are not included in the denominator. Merged-cell handling, header/data-row labelling, footnote-marker treatment, and repeated row indices are resolved upstream during the cleaning stage (section 2.2) that produces the `*_cleaned` inputs, not by the comparison script itself; the comparison operates on whatever string is present in each cleaned cell. Blank cells and “-”/“..”/“...” placeholders are retained and compared as literal strings. A cell counts toward the primary numeric denominator in table 1 ($n=28,255$) if the reference (manual) value consists only of digits with optional thousands separators, regardless of what Textract produced for that cell. A cell counts toward the secondary, numeric-both-sources denominator ($n=26,069$) only if the Textract value is also numeric by the same rule; cells where either side is empty or a placeholder are excluded from both numeric denominators. The row-shift tolerance (table 1, Stage 2) checks whether a Textract cell matches the reference either in its original column or in the adjacent column one position to the right.

Cells excluded by the overlap restriction. Because the comparison in table 1 scores only the overlapping row/column region of each weekly table pair (the inclusion rule above), it does not score every cell Textract produced. Across the 52 cleaned 1956 tables, the evaluated overlap totals 53,872 cells, while the full cleaned Textract bounding grids (`data/textract_cleaned/1956`, summed row $\times$ column extent per file) total 61,116 cells – so 7,244 Textract cells (11.9% of the full grid) fall outside the evaluated overlap and are not scored by any of the agreement figures in this paper. These excluded cells are not automatically 7,244 additional errors: they can include border/padding rows introduced by the cleaning step, repeated header rows, or other structural artefacts of aligning two independently formatted tables, rather than substantive disease-count cells. We report this count for transparency but have not manually audited the 7,244 excluded cells to determine how many are non-trivial content versus padding; this is a scope limitation of the reported accuracy figures (see section 6.2).

4.1.2 String Differences and Levenshtein Distance

To quantify discrepancies between cells with extracted string and their manually entered counterparts, we compute string-level similarity metrics, with a primary focus on Levenshtein distance [14]. Levenshtein distance measures the minimum number of single-character edit operations – insertions, deletions, or substitutions – required to transform one string into another; a distance of zero indicates an exact match, while larger values reflect increasing divergence. The standard dynamic-programming recurrence for computing it is given in Supplementary Material (S4).

The dynamic-programming formulation above follows the general string-to-string correction framework of Wagner and Fischer [15]. In practice, Levenshtein distance is often normalized by string length to enable fair comparison across tokens of different sizes. This is particularly important when evaluating OCR output, where short strings (e.g., years or counts) are disproportionately sensitive to single-character errors.

4.2 Epidemiologically Significant Statistics Derived from Textract Data

4.2.1 Row and Column Sums

Aggregate counts obtained by summing across diseases or locations are often needed, and where the original documents provide these sums, they offer a valuable consistency check: summing values across a row or down a column and confirming the result matches the reported total flags entries that were missed, duplicated, or mistyped. Errors in both the extracted sums and incidence values could in principle cancel out, but this is unlikely; discrepancies flag rows or columns for closer review, letting errors be identified and corrected before downstream analysis.

4.2.2 Cross-Correlation Between Diseases

Computing cross-correlation between two disease time series measures how strongly their patterns move together over time, and whether changes in one tend to occur before, after, or alongside changes in the other – e.g. whether Measles peaks are followed by Chickenpox peaks one or two weeks later. Rather than asking whether two diseases are correlated in a broad sense, cross-correlation examines their relationship across different time shifts, comparing the standardized values of one series with shifted versions of the other to produce a correlation value at each lag.

Ordinary correlation measures the linear association between two random variables X and Y as the normalized covariance, a quantity in $[-1, 1]$ giving the strength and direction of linear dependence. For time series $\{x_t\}$ and $\{y_t\}$, ordinary correlation compares values at the same time index:

$$\rho_{xy}(0) = \frac{\text{Cov}(x_t, y_t)}{\sigma_x \sigma_y}.$$

Cross-correlation extends this idea by introducing a time shift, called the lag. The cross-

240 covariance at lag k is

$$\gamma_{xy}(k) = \text{Cov}(x_t, y_{t-k}),$$

241 and the cross-correlation function (CCF) is

$$\rho_{xy}(k) = \frac{\gamma_{xy}(k)}{\sigma_x \sigma_y}.$$

242 Thus $\rho_{xy}(k)$ is the correlation between x_t and a shifted version of y_t ; varying k measures
243 linear dependence across different time offsets.

244 **Interpretation:**

- 245 • $k = 0$: simultaneous association.
- 246 • $k > 0$: x leads y by k time units.
- 247 • $k < 0$: y leads x by $|k|$ time units.

248 We compute the CCF, $\rho_{xy}(k)$, for x and y referring to two different diseases reported in
249 the same period – once from the manually curated data and once from the automatically
250 extracted data. If the two resulting CCFs are sufficiently similar, this supports the automatic
251 extraction being accurate enough for epidemiologically meaningful inferences.

252 We assess similarity across all lags k by comparing (i) the lag of maximum correlation, (ii)
253 the magnitude of peak correlations, and (iii) the overall shape of the CCF curves, quantified
254 by the difference

$$\Delta(k) = \rho_{xy}^T(k) - \rho_{xy}^M(k),$$

255 where superscripts M and T denote manual and Textract data. Similarity in magnitude,
256 peak location, and overall structure suggests the extracted data preserves the dependence
257 structure of the manual dataset.

4.2.3 Other Time Series Statistics

Another verification tactic is to filter the data by disease – or more generally, by category or type – and examine each subset independently; this makes it easier to spot values inconsistent with the typical range for that group, with outliers identified by scatterplots, boxplots, interquartile range, or standard-deviation cutoffs. Using more than one approach reduces the risk of missing meaningful anomalies or over-flagging valid but extreme observations, and helps distinguish true data-entry errors from legitimate but rare values.

Of the statistics an epidemiologist would naturally want, the *peak height* (the largest observed value) and *peak time* (when it occurs) help predict the intensity of an outbreak.

Peak-detection rule (tables 4 to 5). Because a raw weekly maximum is sensitive to isolated OCR spikes and single-week noise, peaks are located from a locally smoothed series rather than the raw weekly values: within each fitting window, we fit a local quadratic regression (LOESS, span 0.4, degree 2) to weekly cases against week index, and report the week and fitted value at which this smoothed curve is maximized. Ties in the smoothed maximum resolve to the earliest week in the window. Weeks with a missing time index or missing case count are dropped prior to smoothing rather than imputed; a window is not fit if fewer than 5 non-missing weeks remain. We separately checked whether any reported peak week (Measles: weeks 64 and 126; Chickenpox: weeks 64, 112, and 114) coincides with a week flagged by the row-sum validation of section 5.3: none can be checked, because that validation covers only the 52 weeks of 1956, while every reported peak here falls in 1957 or 1958. This is itself a limitation of the current validation coverage, not evidence that the reported peaks are error-free.

Testing the growth rate is also important, as it is a preliminary statistic researchers want to know about an epidemic: estimating the initial growth rate of a disease time series quantifies how quickly cases increase during the early phase of an outbreak, giving a concise measure of epidemic speed that supports comparison across locations or time periods. The

early phase of an outbreak is typically well-approximated by exponential growth, where the expected number of cases increases proportionally to the current number of cases. The initial exponential growth rate, denoted r throughout this paper, is particularly important as it yields the epidemic doubling time ($\frac{\ln 2}{r}$); in principle it can also derive quantities such as the basic reproduction number $\mathcal{R}_0$ [16], though we do not compute $\mathcal{R}_0$ here – we mention it only as context for why r is of independent epidemiological interest, not only an intermediate step toward the doubling time.

Methods for estimating the initial exponential growth rate [17, 18] typically exploit the fact that if incidence grows exponentially, cumulative incidence does too. Since datasets are often observed at substantial intervals (e.g., weekly), methods using “interval incidence” (the difference between cumulative incidence at observation times) are more robust. The growth rate also gives the doubling time, a statistic that is easier to interpret numerically: The exponential growth rate r is related to the doubling time T_d by

$$T_d = \frac{\ln 2}{r}$$

Model specification (tables 4 to 5, figs. 15 to 16). Growth rates and doubling times for Measles and Chickenpox were estimated with `epigrowthfit` 0.15.5 [18] in R 4.5.1, using `egf_model(curve = "logistic")` with all other model arguments left at their package defaults (negative-binomial dispersion, no day-of-week effect – not applicable, since our data are weekly rather than daily – and no excess-cases term). Every top-level model parameter, including $\log(r)$, used the default intercept-only formula, so a single growth rate r was estimated jointly from both fitting windows for a given disease and data source, rather than a separate rate per window; section 5 reports what this implies for the two-window presentation in figs. 15 to 16. Confidence intervals are Wald intervals computed on the $\log(r)$ scale via `confint.egf()` and back-transformed. The two fitting windows (week 35–64 and week 95–126, bracketing the two annual epidemic waves visible in the incidence curves) were

chosen by visual inspection of the manually entered time series and then applied unchanged to the corresponding Textract series for the same disease; windows were not re-chosen from the Textract data itself. Weeks with a missing time index or case count inside a window are dropped prior to fitting, matching the treatment used for peak detection above.

4.3 Use of AI Tools

We used Anthropic’s Claude (Claude Opus 5.5, via `claude.ai` and Claude Code) and OpenAI’s ChatGPT (GPT-4o and GPT-5) to assist with Python and R coding, $\text{\LaTeX}$ formatting, checking consistency between the manuscript text and analysis outputs, and language editing. All AI-assisted code was reviewed by the authors, and all reported results were regenerated from the scripts in the project repository. All scientific arguments, analyses, interpretations, and conclusions are the authors’ own, and the authors take full responsibility for the final manuscript. ChatGPT was also used in a brief exploratory transcription pilot described in section 6.1 and Supplementary Material (S5 Text); no results from it are used in the analyses.

5 Analyzing the results from the Canadian Disease Incidence Dataset

This section explores the accuracy of table extraction on the Canadian Disease Incidence Dataset (CANDID).

5.1 Comparison to Manually Entered Data

5.1.1 Simple Metrics

The cell-agreement analysis in this section covers the 1956 CANDID tables only, not the full 1956–1958 (156-table) dataset described in section 3.1. It was computed with the evaluation

script in `postprocessing/combine.ipynb`, run over the 52 weekly tables for 1956 (53,872 evaluated cells in total; 26,069 of these hold numeric values on both the Textract and reference sides). When comparing values cell by cell, the percentage of cells between the raw Textract data and the manually entered reference transcription that were equivalent (both content and position, i.e. strict string equality; see section 4) was 74.0% across all evaluated cells, or 71.9% among the 28,255 cells numeric in the manual reference (77.9% if instead restricted to the 26,069 cells numeric on both sides). Applying a sequence of processing stages to the Textract output increased content-only agreement (tolerant of comma/whitespace formatting and of a value being shifted into an adjacent column) to 88.5% across all cells, and to 92.9% among cells numeric in the manual reference – the primary numeric-accuracy figure used in this paper, since it does not exclude cells where Textract failed to extract any number at all (a secondary figure of 95.6% obtains under the numeric-on-both-sides denominator). Table 1 summarizes each stage, its resulting agreement under all three denominators, and, critically, whether the underlying rule was fixed independently of this dataset (*a priori*), developed by inspecting the reference transcription, or applied as a manual, post-hoc correction. This distinction matters because rules developed by inspecting the reference data are not validated against unseen material (see the leakage discussion in section 6.2).

5.1.2 String Differences

Using Python’s NLTK library, we compute the Levenshtein distance between extracted and reference strings to find the distribution of edit distances among mismatched entries, from cells differing by a single character to cells with little resemblance to the reference value, and the average distance among entries with an actual error. Recall from section 5.1.1 that content-only agreement is 88.5% across all evaluated cells, distinct from the strict content-and-position agreement of 74.0%, in which a cell’s content may be correct but extracted to the wrong location. Below, “mismatched” cells are those failing even the content-only criterion (`Any_Equality = False`), i.e. the residual disagreements not resolved by any processing

Stage	Operation	All cells (n=53,872)	Numeric, manual ref. (n=28,255)	Numeric, both sources (n=26,069)	Rule provenance
0	Raw Textract output (strict equality)	74.0%	71.9%	77.9%	—
1	+ Comma/whitespace normalization	76.8%	77.3%	83.8%	generic, a priori
2	+ Row-shift tolerance (adjacent column)	88.5%	92.9%	95.6%	developed against ref- erence data
3	Manual correction following inspection (Meningitis, Mumps)	not quantified	not quantified	not quantified	manual, post-hoc

Table 1: Pipeline stages applied to the raw Textract output for the 1956 CAN-DID tables, and the resulting cell-agreement percentage at each stage under three denominators, as computed by `postprocessing/combine.ipynb` and exported to `data/analysis_intermediates/accuracy_summary.csv`. Stage 2 reports content-only agreement (a match either in place or shifted one column over); it is not a strict content-and-position metric. The primary numeric-accuracy figure used elsewhere in this paper is the “Numeric, manual reference” column (n=28,255: every cell where the manual transcription holds a numeric value, regardless of what Textract read there), since it does not silently exclude cells where Textract failed to extract a number at all. The “Numeric, both sources” column (n=26,069) is reported as a secondary figure for comparability with denominators used elsewhere in the OCR-evaluation literature.

stage in table 1. The average Levenshtein distance among these cells is approximately 1.84; the modal distance is 2 (51.7% of mismatched cells), with 1 the next most common (38.1%). fig. 2 shows the proportion of cell contents per Levenshtein distance.

The most common Levenshtein distances are 1 or 2, as expected, consistent with section 5.1.1’s finding that most cell contents differ by a single digit or special character. Of the 1956 tables, 6,177 of 53,872 evaluated cells (11.5%) are true extraction errors (mistakes not resolved through post-processing); of these, 38.1% have a Levenshtein distance of 1 and 51.7% have a distance of 2.

The distance drops off steeply after 2, with a thin tail out to a maximum of 25 – likely cases where the OCR engine confuses a blurry part of a number, or the table structure or scan quality precludes accurate extraction of one or two digits. Table 2 gives the percentage of each Levenshtein distance in cells with errors from the 1956 tables.

The outliers at distances 8, 9, 11, 14, and 25 are almost certainly specific to this dataset.

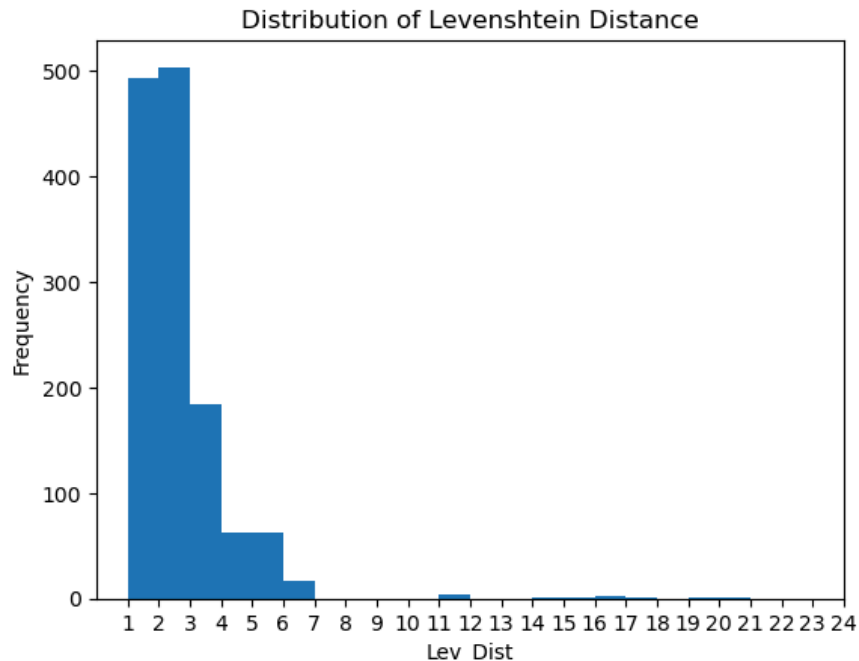


Figure 2: Frequencies of Levenshtein distances (section 4.1.2) between manual and Textract cell values in CANDID; most distances are 1 or 2 characters.

For example, the single distance-25 outlier is a cell where Textract returned a run of placeholder dots (.....) against the reference value *68*: the entire reference string had to be inserted character-by-character. Longer strings, or cells where Textract extracts no numeric content at all, contribute to larger distances.

For distances greater than 6, the errors we inspected occur in the disease-name column (column index 0), where Textract extracted a word or word-fragment (e.g. “meningococcal”, “infectious”, “epidemic”, “primary”) into a cell the reference transcription records as a plain numeric value or placeholder – consistent with row misalignment, where a multi-word or multi-row disease-name label bleeds into the row Textract assigned to the adjacent numeric entry. Such large-magnitude distances are therefore concentrated in disease-name strings and row-alignment failures rather than numeric values, and are clear to a human on inspection.

We also examined the relative Levenshtein distance (distance proportional to the length of the reference string, fig. 3), which accounts for longer strings having a higher potential distance.

Levenshtein Distance	Count	Percentage
1	2356	38.14%
2	3194	51.71%
3	314	5.08%
4	98	1.59%
5	114	1.85%
6	84	1.36%
8	3	0.05%
9	2	0.03%
11	7	0.11%
14	4	0.06%
25	1	0.02%

Table 2: Distribution of Levenshtein distances between extracted and reference values, computed by `postprocessing/combine.ipynb` over the 1956 CANDID tables. Counts sum to 6,177 cells with true extraction errors, out of 53,872 evaluated cells.

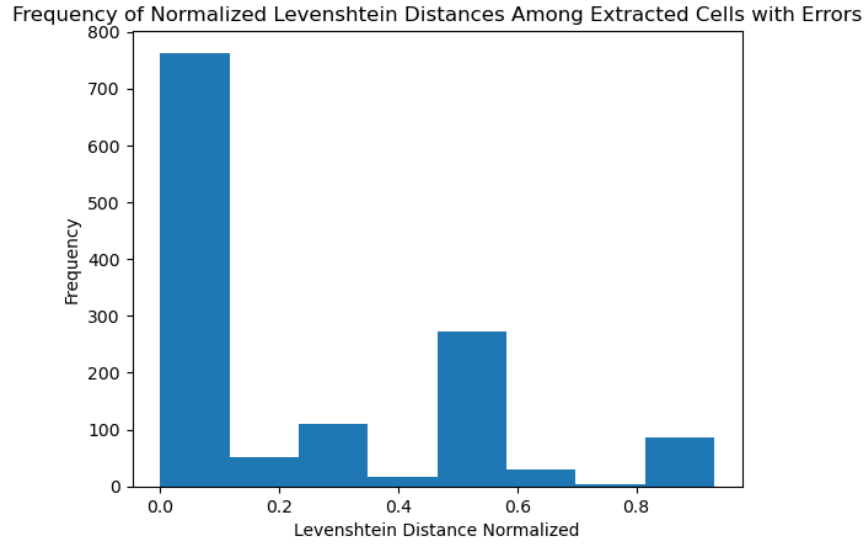


Figure 3: Levenshtein distance (section 4.1.2) normalized by the length of the manually entered cell value.

The most common normalized Levenshtein distance is 0.1. Spikes around 0.5 and above 0.9 are attributable to shifted cells, e.g. where Textract places a disease name into a cell expected to hold a disease count, producing very different character counts.

5.1.3 Detecting Minor Errors with Row Sums

To mitigate errors from row misalignment during extraction, rows corresponding to the same disease were first grouped from each table. Because similar shifting occurs for the same disease across tables, grouping by disease before checking row sums enforces consistent indexing and ensures quantities are evaluated over the correct entries. Figure 4 illustrates the data structure: the week’s total incidence is in a column titled “report week” under “Canada,” with contributing cell values appearing every third column for each province and territory.

DOMINION BUREAU OF STATISTICS																													
HEALTH AND WELFARE DIVISION																													
Summary of Cases of Notifiable Diseases in Canada, as reported by the Provincial Health Departments																													
for the week ended March 2, 1966, in comparison with the previous week, and the 5-year (1961-5) Median																													
DISEASE	CANADA ¹			S.F.B.			P.E.I.			N.S.			C.B.			ONT.			MAN.			SASK.			ALTA.			B.C.	
	Report Week 1966	Previous Week 1966	Cumulative Total ³ 1961-5	Report Week 1966	Previous Week 1966	Median 1961-5	Report Week 1966	Previous Week 1966	Median 1961-5	Report Week 1966	Previous Week 1966	Median 1961-5	Report Week 1966	Previous Week 1966	Median 1961-5	Report Week 1966	Previous Week 1966	Median 1961-5	Report Week 1966	Previous Week 1966	Median 1961-5	Report Week 1966	Previous Week 1966	Median 1961-5	Report Week 1966	Previous Week 1966	Median 1961-5		
1 Chickenpox	1,824	1,177	11,132	1,132	10,139	13,139	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
2 Diphtheria	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
3 Dysentery ⁴	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13	13		
4 (a) Amoebic	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
5 (b) Bacillary	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11		
6 Cerebralitis, infectious	1,072	1,072	6,536	13,152	1,892	1,892	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
7 Influenza, epidemic	1,253	1,247	1,654	15,603	14,385	16,338	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3		
8 Measles	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7		
9 Meningitis, meningococcal	274	1,102	1,741	5,305	7,342	9,027	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
10 Strep	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5		
11 Poliomyelitis, epidemic ⁵	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
12 (a) With paralysis	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
13 (b) Without paralysis	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
14 Rubella	2,072	1,428	5,477	11,831	3,446	2,550	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
15 Scarlet fever, etc. ⁶	402	342	552	2,300	2,112	3,472	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
16 Smallpox	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
17 Tuberculosis ⁷	203	156	515	1,551	1,554	1,754	24	2	2	1	1	1	2	14	12	8	97	54	75	36	17	85	13	11	14	5	8		
18 (a) Pulmonary	167	120	167	1,481	1,131	1,454	24	2	2	1	1	1	2	14	12	8	97	54	75	36	17	85	13	11	14	5	8		
19 (b) Non-pulmonary	10	13	2	70	104	85	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
20 Typhoid and paratyphoid	5	4	7	64	51	30	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
21 Evident fever	1	4	5	29	13	20	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
22 Venereal diseases:	350	303	345	2,922	3,204	3,213	7	8	11	1	1	1	0	11	16	4	5	109	60	82	73	46	87	43	33	22	10		
23 (a) Syphilis, primary	4	1	2	25	25	40	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
24 (b) "secondary	1	3	12	5	26	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
25 (c) "other	56	27	64	353	407	513	2	1	2	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
26 (d) Gonorrhea	305	274	272	2,632	2,714	2,621	2	7	5	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
27 (e) Other venereal diseases ⁸	2	1	3	3	3	3	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		
28 Whooping cough	197	143	140	1,971	3,390	1,790	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2		

(1) Figures adjusted for delayed reports.

(2) Exclusive of Yukon and Northwest Territories.

(3) Total accumulated to the end of the current week from the first week reported in January of the year stated.

(4) Includes cases where type was not specified.

(5) Includes cases of epidemic poliomyelitis.

(6) Includes cases where type was not specified.

(7) Total accumulated to the end of the current week from the first week reported in January of the year stated.

(8) Includes cases where type was not specified.

Figure 4: Rows contributing to the computed sums are in blue, and the expected sums are in pink, of the tables in CANDID section 3.1. In the original PDF table, we expect that across a row, the sum of each value in the blue columns equals the total in pink.

For each disease, we plot the difference between Textract’s extracted total and the sum calculated by looping over the row contents; fig. 5 shows the values for Chickenpox.

Most rows show zero difference, indicating agreement between extracted totals and re-computed values; nonzero deviations trace to numerical errors in individual cells.

For example, the first outlier in fig. 5 is Week 9. Inspecting the scan, the true total is 1224, from the non-empty contributing values:

55, 495, 379, 25, 270

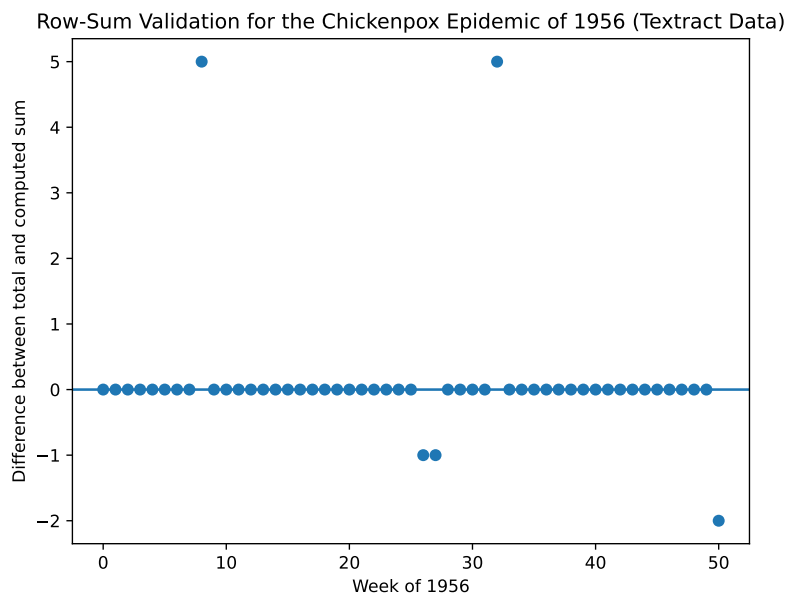


Figure 5: Differences in expected versus calculated row sums for Chickenpox in CANDID (section 3.1) to verify accuracy of cell values (method described in section 4).

Textract correctly extracts the total as *1224*, but differs from the reference on the contributing values themselves:

50, 495, 379, 25, 270

Our algorithm sums these to 1,219, a difference of 5 from the expected total (fig. 5): the value 55 was misread as 50. Such minor numerical discrepancies can thus be found and corrected.

Some discrepancies are more subtle. The first outlier in fig. 7 is also Week 9. The recomputed total from the extracted contributing values is:

5, 13, 59, 782, 802, 61, 2, 47, 111

.

Textract reports a total of *1852*, while the sum of the contributing values is *1882* – a

409 difference of -30 (fig. 7), indicating an error in either the contributing cells or the row total,
 410 or both. The reference transcription records the row total as *1862* (fig. 6), only 10 away
 411 from Textract’s *1852* – not enough to explain the original 30-count gap, leaving a further
 412 20-count gap between the reference total and the sum of its contributing values.

413 Closer inspection of the source image shows the correct total is *1882* (exactly the sum of
 414 the contributing values) – an error in the reference transcription, not the Textract extraction,
 415 arising from scan ambiguity. This is a substantive result in its own right: row-sum validation
 416 detected an error in the reference transcription, showing that internal consistency checks can
 417 adjudicate between the two sources rather than merely score the extraction tool against an
 418 assumed-correct reference (section 6.2).

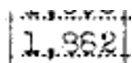


Figure 6: Total Measles instances in Week 9 of CANDID. This is hard to interpret for both OCR and humans.

419 These examples illustrate an advantage of row-sum validation: comparing independently
 420 computed totals against reported values flags anomalies in the dataset and can be adapted
 421 to detect OCR errors generally.

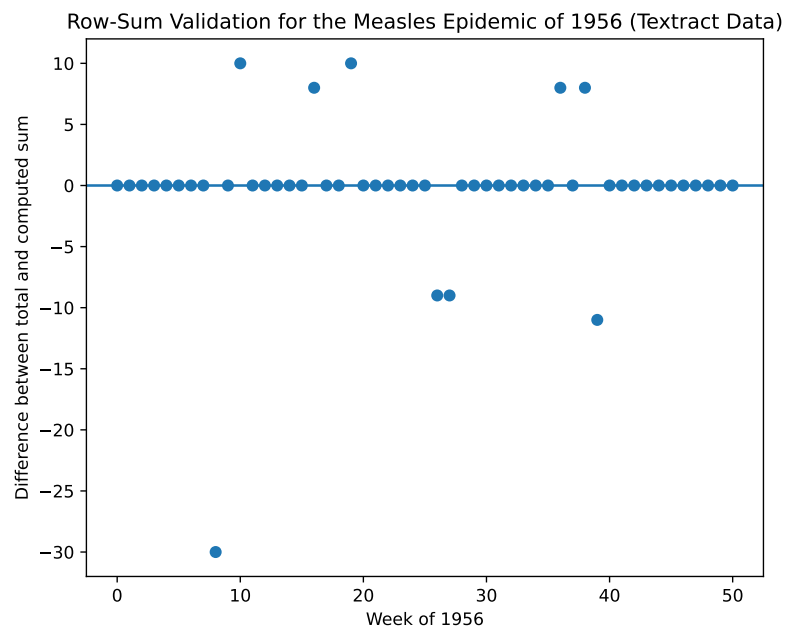


Figure 7: Differences in expected versus calculated row sums for Measles in CANDID (section 3.1) to verify accuracy of cell values (method described in section 4).

5.2 Comparing Important Statistics

5.2.1 Cross-Correlation Between Diseases

Unless otherwise stated, the Textract series used in this subsection are the processed data described in table 1 (Stage 2), without manual correction. The cross-correlation analysis first focuses on Chickenpox and Measles: we extract the relevant time series from both the manually curated and Textract data, then post-process to align formats, handle missing values, and ensure consistent indexing across sources (section 2.2).

Figure 8 plots both time series for an initial qualitative assessment. Visually, the two series are highly similar, indicating the extraction process preserves the overall timing structure.

This is consistent with section 5.1.1’s content-only agreement of 92.9% for cells numeric in the manual reference (95.6% under the numeric-on-both-sides denominator), with most remaining errors involving only small numerical deviations: the dominant trends, seasonal patterns, and relative magnitudes of the two diseases are largely unaffected by extraction noise. The remaining disagreements do change the graphs slightly – a sharp decrease in Chickenpox around week 60, and missing Measles data around week 145 – unsurprising given that the Textract data’s errors are systematic and sharp; for example, the Scarlet Fever comma-induced shift (section 5.3) propagates through all affected rows rather than occurring at random.

The two series rise and fall in roughly the same seasonal pattern, with major peaks and troughs at approximately the same times, so from the incidence curves alone we would expect a fairly strong positive cross-correlation near lag 0.

What matters here is the difference between the cross-correlation computed from the manual versus Textract data. Figure 9 shows both; the quantified comparison follows in table 3.

To gauge similarity, we computed summary statistics across all lags (table 3; per-lag val-

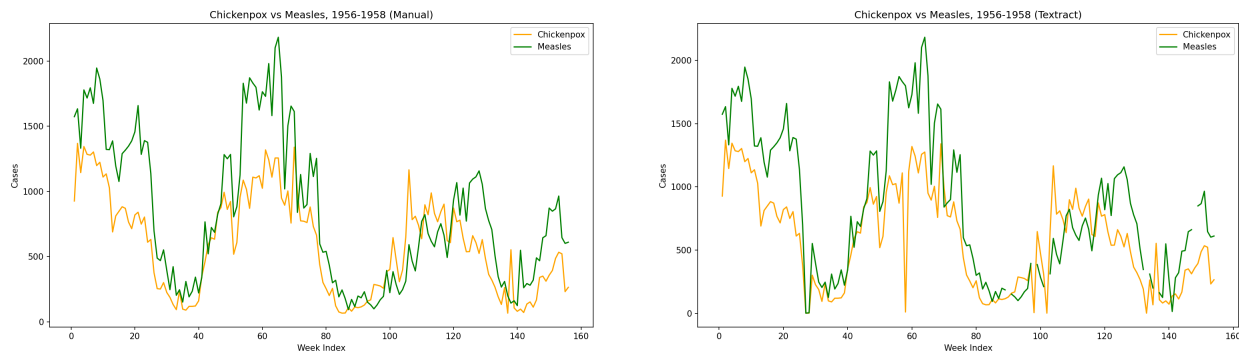


Figure 8: Comparison of manual versus Texttract disease incidence for Chickenpox and Measles. Data version: Texttract series processed through the pipeline in table 1 (Stage 2), without manual correction.

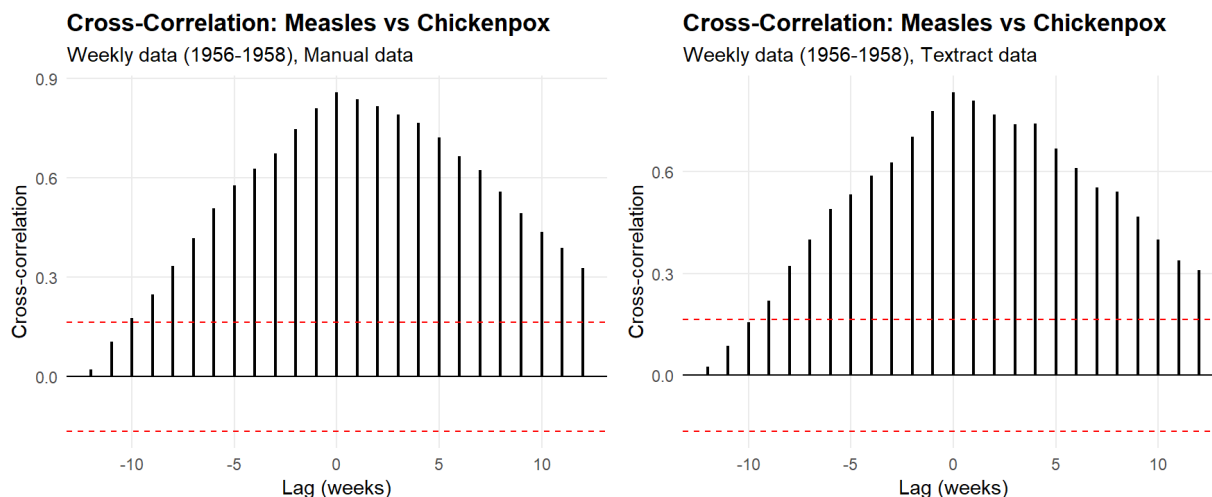


Figure 9: Comparison of manual versus Texttract disease cross-correlation for Chickenpox and Measles. This is described in section 4. Data version: processed Texttract data (table 1, Stage 2), without manual correction. Regenerated from the corrected common-calendar-week CCF calculation (`analysis/cross_correlation.R`); see table 3.

ues in Supplementary Material, Table S1), reporting absolute and root-mean-square differences rather than a relative difference at each lag, since a relative difference is uninformative wherever the manual correlation is near zero, as the next comparison illustrates.

Both datasets peak at lag 0 (table 3), so the change in peak lag is 0 and the change in peak correlation is small (-0.0255). Across all 25 lags, the largest absolute discrepancy is 0.0699 (at lag 7) and the root-mean-square difference is 0.0376. Because cross-correlation measures time relationships rather than exact magnitudes, the estimated timing structure

here was similar between the two sources.

We repeat this for Measles and Meningitis (incidence curves in fig. 10, cross-correlation in fig. 12; summary statistics in table 3, per-lag values in Supplementary Material, Table S2). Unlike the Measles/Chickenpox comparison, the Meningitis series here has been manually corrected: additional post-processing removed visible outliers (section 5.3).

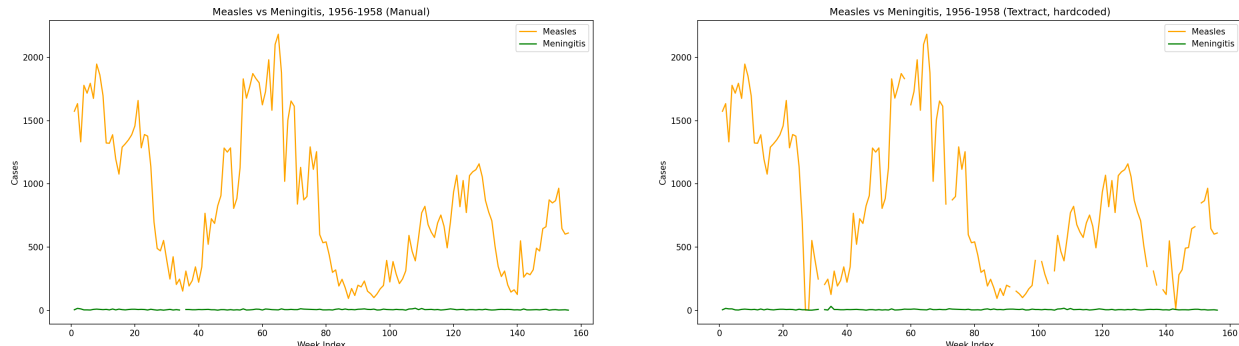


Figure 10: Comparison of manual versus Textract disease incidence for Measles and Meningitis. Data version: manually corrected (outlier removal, section 5.3). The Meningitis data without this additional step is seen in fig. 11.

The cross-correlation between Measles and Meningitis is weak across all lags in both datasets, generally between about -0.11 and 0.09 (table 3). Because these correlations are close to zero throughout, this pair is the one for which a relative-difference metric is least informative: at several lags the manual correlation itself is near zero, so any ratio against it is unstable regardless of how small the absolute discrepancy is. The manual series peaks at lag 3, Textract at lag 2 (a change of -1 week); the change in peak correlation is not small in relative terms, though both values are close to zero (0.0848 manual vs. 0.1366 Textract, a difference of 0.0517). The largest absolute discrepancy is 0.0750 (at lag 4) and the root-mean-square difference is 0.0401 – both larger in absolute terms than the Measles/Chickenpox discrepancies above, and a relative-difference expression of this same gap would read an order of magnitude larger still, for the reason given above. Textract captures the overall pattern here, with the largest discrepancies at lags where both correlations are already small.

Finally, we repeat the procedure for Chickenpox and Mumps. The Textract series here

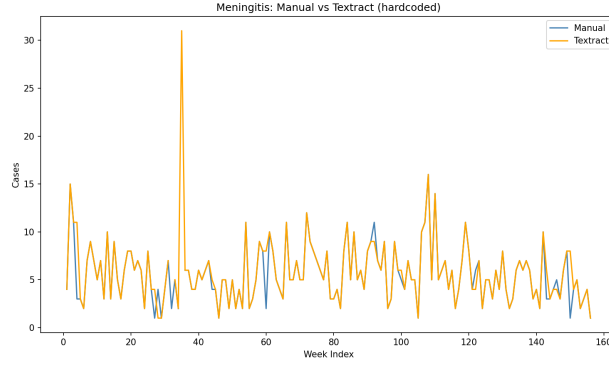


Figure 11: Manual (blue) and Textract (orange) time series for Meningitis. The Textract data shown here is plotted *after* the additional post-processing seen in section 5.3; see table 3 for the quantified comparison.

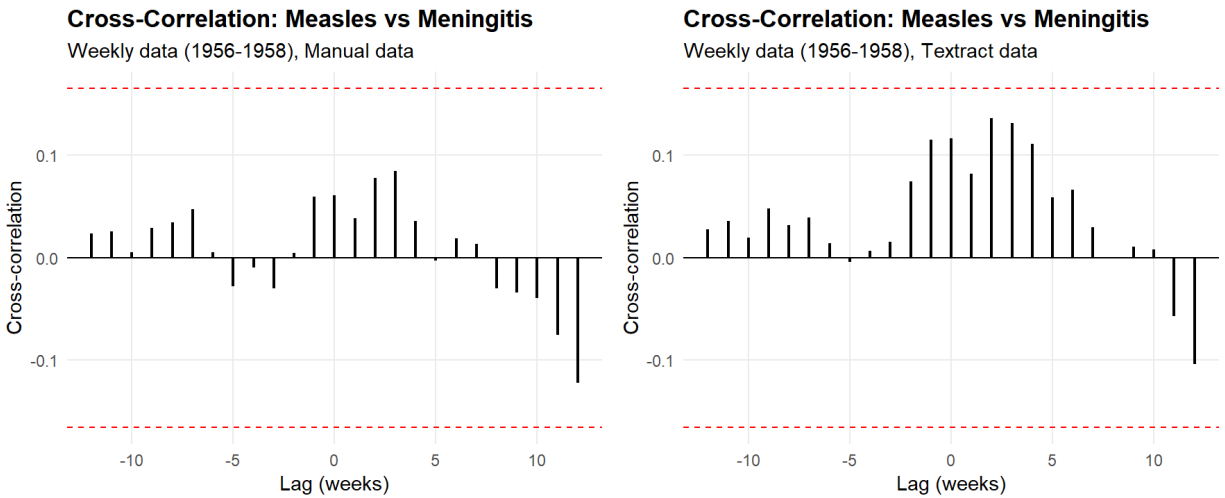


Figure 12: Comparison of manual versus Textract disease cross-correlation for Measles and Meningitis. Data version: manually corrected (outlier removal, section 5.3). Regenerated from the corrected common-calendar-week CCF calculation (`analysis/cross_correlation.R`); see table 3.

are manually corrected, following the same procedure as the Meningitis data (section 5.3): OCR-related outliers are corrected before computing the cross-correlation, so the resulting time series (incidence curves in fig. 13) are suitable for analysis.

Figure 14 shows the cross-correlation between Chickenpox and Mumps is very similar between manual and Textract data; the summary statistics in table 3 (per-lag values in Supplementary Material, Table S3) confirm the differences at each lag are small.

The CCF-based conclusions for this pair are almost identical between manual and Tex-

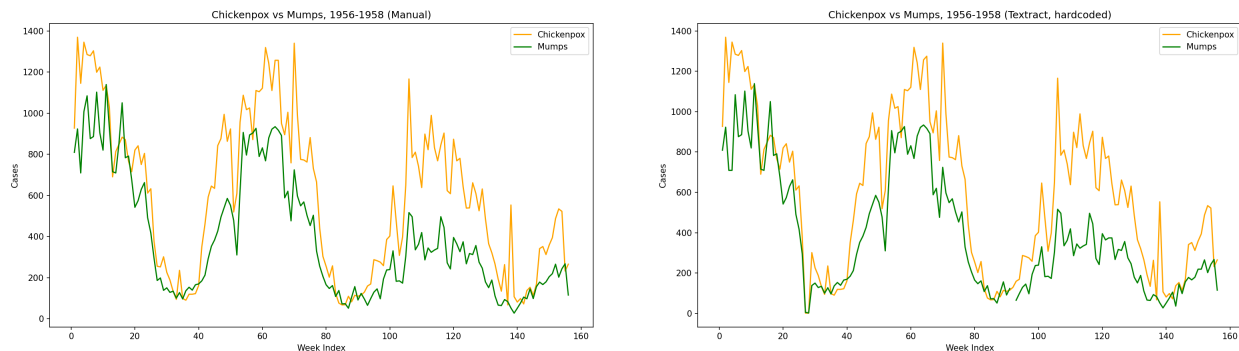


Figure 13: Comparison of manual versus Textract disease incidence for Chickenpox and Mumps. Data version: manually corrected (section 5.3).

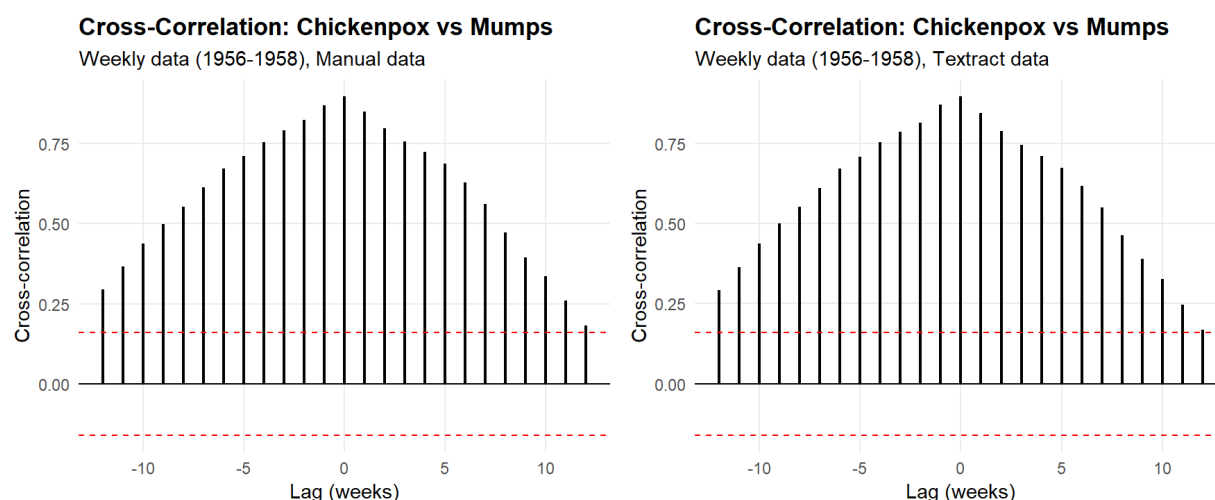


Figure 14: Comparison of manual versus Textract disease cross-correlation for Chickenpox and Mumps. Data version: manually corrected (section 5.3). Regenerated from the corrected common-calendar-week CCF calculation (`analysis/cross_correlation.R`); see table 3.

tract data: the peak lag, most important for epidemiological study, is identical (lag 0 in both), the change in peak correlation is negligible (-0.0009), and both the largest absolute discrepancy (0.0145 , at lag 5) and the root-mean-square difference (0.0080) are the smallest of the three disease pairs examined.

Table 3 collects these summary statistics for all three disease pairs. A strong cross-correlation near lag zero is expected for two seasonal childhood infections circulating in the same population and is not, on its own, a demanding evaluation target; what these comparisons test is whether Textract preserves the *structure* of that dependence (peak lag, and

the overall shape of the CCF) rather than establishing a stringent benchmark of extraction accuracy.

Disease pair	Data version	Max $ \Delta $	Lag at max	RMS diff
Measles vs Chickenpox	processed, no manual correction	0.0699	7	0.0376
Measles vs Meningitis	manually corrected	0.0750	4	0.0401
Chickenpox vs Mumps	manually corrected	0.0145	5	0.0080

Table 3: Summary of cross-correlation agreement between manual and Textract series for all three disease pairs (part 1 of 2; continued in section 5.2.1), computed by `analysis/ccf_summary.py` from the per-lag CCF values (Supplementary Material, Tables S1–S3), which are themselves computed by `analysis/cross_correlation.R` after merging the manual and Textract series on their shared week identifier and restricting to weeks with complete data in both sources at every lag (the two series are no longer NA-dropped independently, which previously let them cover different numbers of calendar weeks per disease pair). Lag range $k \in \{-12, \dots, +12\}$ ($n = 25$ lags). Sign convention: all differences are Textract – Manual. **Data version** follows table 1: two of the three pairs (Meningitis, Mumps) use Textract series manually corrected following inspection (section 5.3); Measles vs. Chickenpox does not.

Disease pair	Lag M	Lag T	Δ Lag	ρ M	ρ T	$\Delta\rho$
Measles vs Chickenpox	0	0	0	0.8584	0.8329	–0.0255
Measles vs Meningitis	3	2	–1	0.0848	0.1366	0.0517
Chickenpox vs Mumps	0	0	0	0.8983	0.8974	–0.0009

Table 3 (continued). Peak lag and peak correlation for each series (part 2 of 2, continued from table 3). “M”/“T” denote Manual/Textract; “Lag M”/“Lag T” and “ ρ M”/“ ρ T” are the peak lag and peak correlation for each series, and the Δ columns are Textract – Manual. Peak-lag ties (smaller $|k|$, then negative lag) did not occur for any of the three pairs.

5.2.2 Time Series Statistics

After isolating the time series for specific diseases in CANDID, we can find the peak heights and times. In fig. 8, the Measles time series (green) from Textract is visually similar to the manually entered data.

We report both the initial exponential growth rate r and the doubling time T_d it implies for the Measles epidemics: r is the quantity the model actually estimates, and T_d is an easier-to-interpret transformation of it, so reporting T_d alone would hide the scale the model was fitted on. Both are calculated with the R package `epigrowthfit` [18], using the model specification in section 4; r is a per-week rate, as our incidence data are indexed in weeks. Figure 15 shows the fitted curves and doubling-time annotation for Measles for both datasets.

Table 4 gives the numeric results comparing growth rate, doubling time, peak week, and smoothed peak incidence between the manually entered and automated datasets.

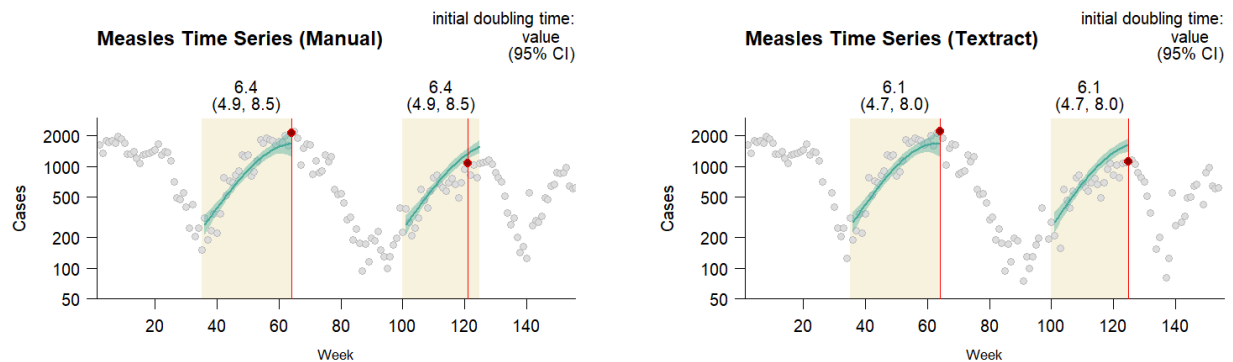


Figure 15: Manual versus Textract Measles growth rates as described in section 4. Data version: processed Textract data (table 1, Stage 2), without manual correction. Shaded bands mark the two fitting windows (week 35–64 and week 95–126). The annotated “initial doubling time: value (95% CI)” is in weeks and is the same number in both shaded windows for a given panel: this is not a labelling error but a direct consequence of the model specification (section 4), which estimates one growth rate shared across both windows rather than a separate rate per window; see table 4 for the corresponding growth rate r .

We can compare these same statistics for a different disease: fig. 16 shows the doubling times and peaks for Chickenpox in the manual and extracted data, compared in Table 5.

Window	Fit range (wk)	n obs	Source	Peak week	Smoothed peak incidence	Δ week	Δ cases (%)	r (wk ⁻¹ , 95% CI)	T_d (wk, 95% CI)
Window 1	35–64	30	Manual	64	1967	–	–	0.137 (0.114, 0.165)	5.1 (4.2, 6.1)
		30	Textract	64	2161	0	+194 (+9.84)	0.124 (0.100, 0.154)	5.6 (4.5, 6.9)
Window 2	95–126	32	Manual	126	1040	–	–	0.137 (0.114, 0.165)	5.1 (4.2, 6.1)
		30	Textract	126	1174	0	+134 (+12.91)	0.124 (0.100, 0.154)	5.6 (4.5, 6.9)

Table 4: Comparison of peak week, smoothed peak incidence, growth rate, and doubling time between Textract and manual data for Measles, recomputed from `data/analysis_intermediates/table4_measles_peak_comparison.csv` and `growth_rate_summary.csv` (analysis/growthrates.R). Data version: processed Textract data (table 1, Stage 2), without manual correction – the only Textract version available for Measles (see section 5). Sign convention: Δ = Textract – Manual. Peaks are located from a LOESS-smoothed series within each fit window (peak-detection rule, section 4); ties resolve to the earliest week. n obs is 30 for every window/source except Manual Window 2 (32); the Textract series is missing 2 weeks (weeks 98 and 102) inside Window 2, which is why the Manual and Textract fits for that window are not run on identical n . r and T_d are estimated from a single growth-rate parameter shared across both fitting windows (model specification, section 4) and are therefore identical for Window 1 and Window 2 within a given source; both fits converged (optim status “relative convergence”). The Manual/Textract 95% CIs for r overlap and each point estimate falls within the other source’s CI (section 5); this comparison is not on strictly identical n given the two missing Textract weeks noted above.

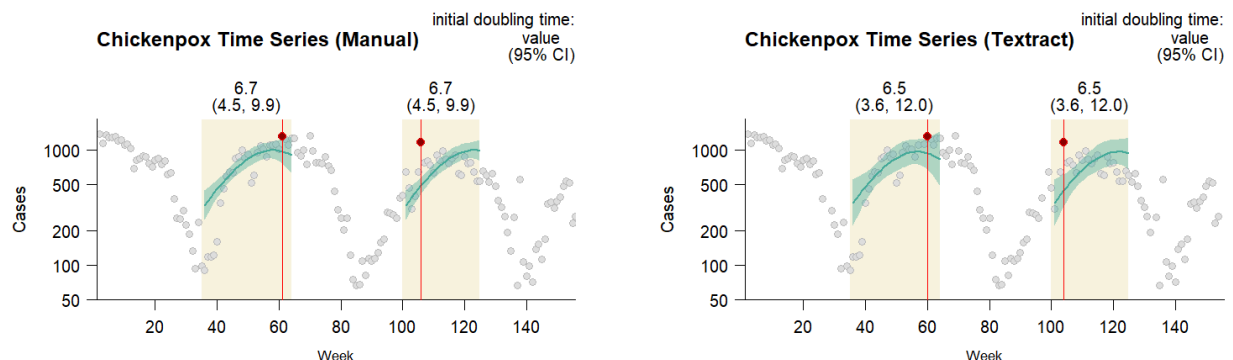


Figure 16: Manual versus Textract Chickenpox growth rates as described in section 4. Data version: processed Textract data (table 1, Stage 2), without manual correction. Shaded bands mark the two fitting windows (week 35–64 and week 95–126). The annotated “initial doubling time: value (95% CI)” is in weeks and is the same number in both shaded windows for a given panel: this is not a labelling error but a direct consequence of the model specification (section 4), which estimates one growth rate shared across both windows rather than a separate rate per window; see table 5 for the corresponding growth rate r .

504 Peak week is unchanged (Textract – Manual = 0) in three of the four windows, and
505 differs by -2 weeks in the fourth (Chickenpox, Window 2); smoothed peak incidence differs

Window	Fit range (wk)	n obs	Source	Peak week	Smoothed peak incidence	Δ week	Δ cases (%)	r (wk ⁻¹ , 95% CI)	T_d (wk, 95% CI)
Window 1	35–64	30	Manual	64	1223	–	–	0.144 (0.125, 0.167)	4.8 (4.2, 5.5)
		30	Texttract	64	1313	0	+90 (+7.42)	0.144 (0.108, 0.193)	4.8 (3.6, 6.4)
Window 2	95–126	32	Manual	114	882	–	–	0.144 (0.125, 0.167)	4.8 (4.2, 5.5)
		32	Texttract	112	882	–2	0 (0.00)	0.144 (0.108, 0.193)	4.8 (3.6, 6.4)

Table 5: Comparison of peak week, smoothed peak incidence, growth rate, and doubling time for Chickenpox between Texttract and manual data, recomputed from `data/analysis_intermediates/table5_chickenpox_peak_comparison.csv` and `growth_rate_summary.csv` (`analysis/growthrates.R`). Data version: processed Texttract data (table 1, Stage 2), without manual correction – the only Texttract version available for Chickenpox (see section 5). Sign convention: Δ = Texttract – Manual. Peaks are located from a LOESS-smoothed series within each fit window (peak-detection rule, section 4); ties resolve to the earliest week. r and T_d are estimated from a single growth-rate parameter shared across both fitting windows and are therefore identical for Window 1 and Window 2 within a given source; both fits converged. The Manual/Texttract 95% CIs for r overlap and each point estimate falls within the other source’s CI (section 5); note the Texttract CI is noticeably wider than the Manual CI (0.108–0.193 vs. 0.125–0.167).

by up to +194 cases (Measles Window 1, +9.84%) in absolute terms and by up to +12.91% (Measles Window 2, +134 cases) in relative terms – the largest absolute and largest relative discrepancies fall in different windows – with Texttract consistently over-counting relative to Manual across all four windows. None of the four reported peak weeks can be cross-checked against the row-sum validation of section 5.3, since that validation covers only 1956 while all four peaks fall in 1957–1958 (section 4).

For the growth rate itself: the Measles estimate is $r = 0.137$ wk⁻¹ (95% CI 0.114–0.165) from the manual series and $r = 0.124$ wk⁻¹ (95% CI 0.100–0.154) from Texttract, a difference of $\Delta r = -0.013$ wk⁻¹; the corresponding doubling times are 5.1 (4.2, 6.1) and 5.6 (4.5, 6.9) weeks, $\Delta T_d = +0.5$ weeks. For Chickenpox, $r = 0.144$ wk⁻¹ from both sources to three significant figures (95% CI 0.125–0.167 manual, 0.108–0.193 Texttract), $\Delta r \approx -0.0004$ wk⁻¹; doubling times are 4.8 (4.2, 5.5) and 4.8 (3.6, 6.4) weeks, $\Delta T_d = +0.01$ weeks. In both diseases the Manual and Texttract 95% CIs for r overlap, and each point estimate falls inside the other source’s interval; the Chickenpox Texttract CI is also visibly wider than its Manual counterpart, expected given no more data points were available to the fit (tables 4 to 5 report n observations per window). Overlapping confidence intervals are reported here as

a descriptive feature of the two fits, not a statistical test of equivalence; this observation is scoped to the two diseases and two epidemic windows examined here, and does not establish that growth-rate estimation from Textract data is reliable in general.

5.3 Remarks on Dataset Specific Coding

Several challenges arose when computing row-wise sums for each disease. The main issue was extraction errors such as row misalignment or mislabeled diseases, which shifted entire rows of values and propagated consistently across tables, making it difficult to associate values with the correct disease. For example, a disease name containing a comma in the original table caused Textract to extract it as two separate cells, shifting the following incidence values; since each table has the same or similar disease names, this problem recurred in every table at that row.

Grouping entries by disease (rather than by week, as extracted) before summation reduced this complexity, limiting the need for manual intervention since most diseases had identical errors across tables. When a shift occurred, the position of the reported total and its contributing values became offset and the indices used to identify them were no longer valid; correct indices were re-identified through manual inspection and the summation inputs adjusted accordingly.

An example in CANDID is Scarlet Fever: persistent discrepancies between reported and recomputed sums, beginning at week 5 of 1956, traced to the disease label changing from “Scarlet Fever” to “Scarlet Fever, etc.” in the original document. The added comma altered the table parsing, shifting all subsequent cell values in the row and propagating the misalignment through extraction. Resolving this required manually identifying the change and recombining the affected cell values.

A further consideration in the row-sum computation is the structure of the original tables: a row index is printed both at the beginning of each row and repeated at the end (see fig. 1), and the province “Report Week” values are not the only numeric-looking entries in the

548 row. Rather than iterating generically across the row and attempting to filter out non-
 549 contributing values at runtime, the row-sum script (`analysis/row_sum_check.py`) avoids
 550 this problem by construction: it reads the Canada report-week total from a fixed column
 551 and sums only the fixed set of columns, identified directly from the table layout in fig. 1,
 552 that hold each province's report-week value (fig. 17). Because the contributing columns
 553 are specified explicitly rather than inferred from cell content, the row index and any other
 554 non-contributing cell are never candidates for inclusion in the sum.

```
TARGET_COL = 2
CONTRIB_COLS = list(range(8, 36, 3)) # 8, 11, 14, ..., 35 -- one
    per province

target = to_int_or_none(row[TARGET_COL])
contributors = [(c, to_int_or_none(row[c])) for c in CONTRIB_COLS]
computed = sum(v for _, v in contributors if v is not None)
```

Figure 17: The column layout used by the row-sum validation script (`analysis/row_sum_check.py`) to identify the Canada report-week total (column 2) and each province's contributing report-week value (columns 8, 11, 14, ..., 35, one per province at three-column intervals), as reproduced from the actual script rather than paraphrased.

555 We also had to correct extraction errors when comparing epidemiological statistics be-
 556 tween the two datasets. The extracted Meningitis time series contains large, isolated spikes
 557 inconsistent with expected disease patterns; because these would distort the cross-correlation
 558 analysis, they were corrected before proceeding. Figure 19 shows the unprocessed Measles
 559 and Meningitis series: Measles follows a smooth, expected pattern, while Meningitis includes
 560 several sharp spikes from OCR misreads.

561 Figure 18 shows another example, the Textract time series for Mumps versus Chickenpox
 562 before manual correction.

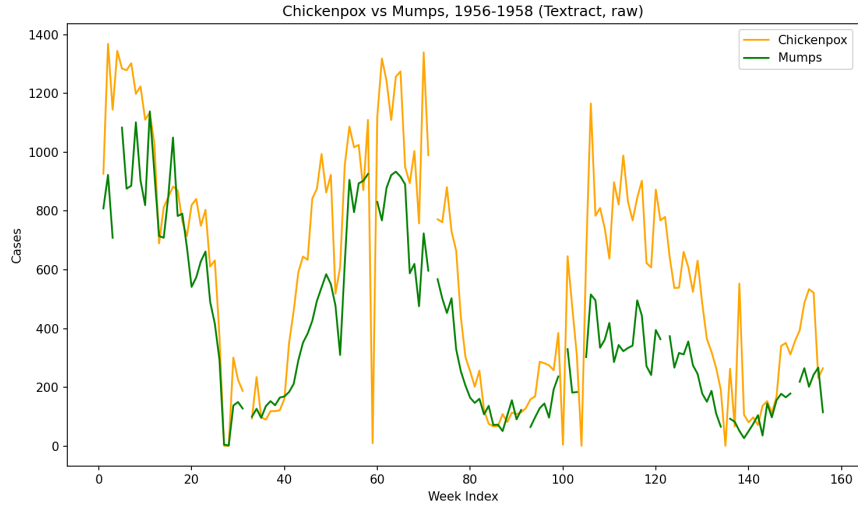


Figure 18: Data version: raw/uncorrected Textract values (prior to the manual correction in section 5.3). The time series for Chickenpox and Mumps from Textract, with clear errors seen around weeks 60, 100, 110, etc as well as missing values.

Although these errors are large, they occur only occasionally: a small number of spikes were identified across the affected time series using a boxplot or violin plot (fig. 20), which highlights extreme values against the rest of the data.

Correcting these incidence values by checking against the original dataset took less than 20 minutes for the Meningitis and Mumps series reviewed here (three years of weekly data). This is a measured correction time for these specific series, not an estimate of total post-processing effort across the full dataset.

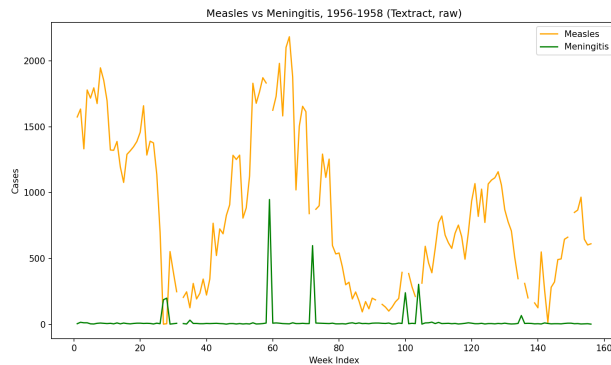


Figure 19: Data version: raw/uncorrected Textract values (prior to the manual correction in section 5.3), showing clear errors in the meningitis cases.

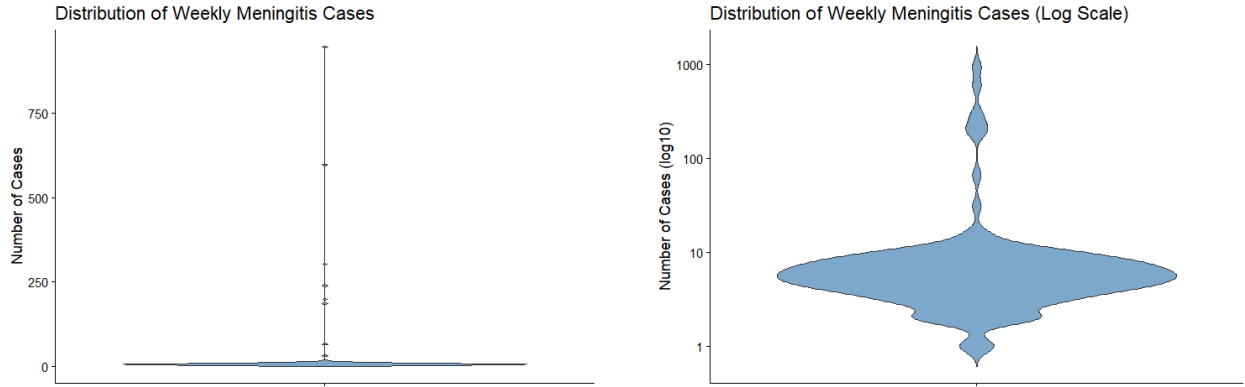


Figure 20: Violin plots of Textract Meningitis data shown on linear (left) and log (right) scales.

6 Discussion

6.1 Impact

For epidemiology researchers, automating tabular data entry is a major aid in studying historical data: it accelerates processing large datasets and frees effort for interpretation rather than transcription. Because historical sources are often vast, messy, and inconsistent, automation makes it possible to standardize and analyze them at scale, revealing patterns that would be implausible to uncover manually in comparable time. Our conclusions are limited to printed, regularly formatted CANDID tables from 1956–1958, processed by the specific workflow described above; we do not claim they generalize to handwritten documents, irregularly formatted tables, or other extraction tools or model versions.

For an automation tool to successfully improve and accelerate data entry, it must produce a dataset that is either close to the original, or easy to process and fix. If too much human correction is needed, the time saved becomes small, and data correction is a less pleasant task than data entry even with modest savings. In our preliminary exploration (Supplementary Material, S5 Text), the conversational model we tried before Textract produced errors tedious enough to fix that we abandoned it for a tool purpose-built for table extraction.

By comparison, Textract’s errors are of a different character than typical manual-entry errors: often either small (one or two characters from the reference, with little influence on downstream statistics) or large. The large-magnitude errors are relatively rare and occur specifically in disease-name string values (section 4.1.2).

Beyond the initial technical hurdle of using AWS Textract programmatically and the unique post-processing each dataset requires, recreating CANDID with Textract appears to take less overall time and effort than manual data entry. Similar tables (in size and complexity) took on average 30 minutes to fully digitize from a scan by a member of McMaster’s Theoretical Biology group; extrapolated at that rate to all 156 tables from 1956–1958, man-

ual entry would take an estimated 78 hours – an extrapolation from one operator’s rate, not a measurement of the full manual process. On the automated side, we directly measured script runtime (≈ 15 minutes) and the correction pass (< 20 minutes for the reviewed series, section 5.3); script development time and remaining post-processing time were not measured. With these caveats, extraction runtime was short relative to the extrapolated manual-entry estimate, but total person-time was not measured.

Because document-extraction systems evolve rapidly, these findings should be understood as an evaluation of one service and workflow on one historical archive, not a recommendation of Textract for new digitisation projects. Researchers should benchmark currently available tools on representative pages and independently audit both table structure and cell contents before selecting a workflow.

6.2 Limitations

No adjudicated reference standard. The manual transcription is used throughout as the comparison basis without independent re-adjudication: there was no double transcription or two-reviewer resolution of discordant cells. The paper itself shows this reference can be wrong – the Week 9 Measles total (section 5), where the source scan supports *1882* and the reference records *1862*. Reported disagreement counts therefore conflate extraction and transcription errors, with the direction of bias unknown; resolving this would require two independent reviewers adjudicating a random validation sample against the source scans, which was not done here.

A further limitation is data leakage in developing our normalization and structural-repair rules: the comma/whitespace normalization, header-exclusion, and adjacent-cell tolerance rules in section 5.1.1 (table 1) were developed or tuned while the reference transcription was visible to us, with no held-out test set. This also applies to the Scarlet Fever comma fix, the fixed-column row-sum computation, and the Meningitis/Mumps spike correction (all section 5.3). Because these rules were shaped by errors observed in this dataset, the reported

agreement figures are an upper bound on expected performance on unseen CANDID tables or other datasets, not an estimate of out-of-sample accuracy.

Evaluation year. Cell-level agreement (section 5.1.1–section 4.1.2) covers 1956 only (52 of 156 tables); downstream epidemiological statistics (section 5) use all three years. Worth naming explicitly: the peak weeks in the growth-rate tables (64, 112, 114, 126) fall entirely outside the range directly measured for cell-level agreement, so their accuracy rests on the untested assumption that agreement in 1957–1958 resembles 1956.

Selection of downstream comparators. Three disease pairs for cross-correlation and two diseases for growth-rate comparison were chosen without a pre-specified selection rule. A strong cross-correlation near lag zero is expected for two seasonal childhood infections in the same population and is not, on its own, a demanding target; what these comparisons test is whether Textract preserves the *structure* of that dependence rather than establishing a stringent accuracy benchmark.

No external evaluation on a structurally different archive. All results come from a single source (CANDID) with a consistent printed template; transfer to other table layouts, languages, typographies, or scan qualities is untested. section 6.3 discusses a structurally distinct Quebec provincial dataset as a candidate for such a test; we have not analysed it here.

Single tool, single version. This study evaluates one extraction tool, not a controlled comparison against alternatives; the informal exploration in the Supporting Information (Supplementary Material, S5 Text) that led us to Textract was not a rigorous benchmark, and no controlled comparison against other document-extraction tools or current document-capable language models was performed. We also used one commercial service at one point in time, without pinning a specific model or API version; Amazon may change Textract’s

underlying models without notice, so results reported here may not reproduce exactly if re-extracted today. Our choice of Textract reflects its suitability for this task, not a demonstrated advantage over alternatives.

The growth-rate comparison (tables 4 to 5) is similarly limited: only the Stage-2 processed series, without manual spike correction, is available for Measles and Chickenpox, since manual correction (section 5.3) was applied only to Meningitis and Mumps. We are therefore unable to test whether the growth-rate estimate is stable across raw, processed, and manually corrected versions of the same disease’s data; doing so would require applying the same manual review to Measles and Chickenpox, which was not done here.

No measured time-and-cost study. The labour comparison in section 6.1 is an extrapolation from one operator’s rate, not a controlled measurement: script development time, dataset-specific coding time, validation time, retries, and AWS charges were not measured, and only extraction runtime (≈ 15 minutes) and one correction pass (< 20 minutes, two series) were timed directly. A defensible comparison would record person-hours by activity, operator count and experience, machine time, itemised AWS charges with a pricing date, and would separate first-project setup cost from the marginal cost of digitizing an additional year of tables.

The limitations above concern the validity and scope of what was measured. The remainder of this section concerns the operational cost of using Textract – not the validity of the results, but relevant to a researcher deciding whether to adopt this workflow.

The main operational limitation is the barrier to entry of AWS software: limited development experience may increase the effort required. Beyond account set-up and writing the extraction script, connecting to an AWS account involves an AWS ID, secret key, secret access key, and more (Supplementary Material, S1). Once set up, though, researchers have complete control over how data is processed and extracted – e.g. omitting confidence levels, or processing only certain table elements or pages – and the same extraction script works

for any scanned table document, unlike the data-specific post-processing that follows.

Another operational limitation is extraction time: extracting the full 156-table CANDID dataset took around 15 minutes. This could likely be reduced, e.g. by cropping scans to the relevant table region or omitting confidence levels, but optimizing API calls was not a priority here.

The largest operational limitation is the time and effort to process and clean data post-extraction: formatting errors (merged cells, shifted rows) and large errors in rare numeric values and string headers together took more time to find and correct than extraction itself. Small errors, present throughout, can be particularly hard to detect when they occur in numerical values that appear plausible on their own, and often do not stand out unless validated against source data.

6.3 Future Work

Future trials on different datasets will reveal more about the power and limitations of tools such as Textract, expanding to more complicated tables and handwritten documents. One natural next test case is a Quebec provincial communicable-disease dataset (Dominion Bureau of Statistics, July 1927–December 1931), collected by McMaster’s Theoretical Biology group but not previously digitized or transcribed. It is structurally distinct from CANDID in ways that would stress-test this workflow: row headers printed diagonally and in French, and a table layout that changes partway through the series from two tables per page to one. We have not analysed this dataset here – doing so properly would require an independent manual transcription of a stratified sample, beyond this paper’s scope – but it is a natural candidate for validating the workflow’s generalizability beyond a single archive.

Our results suggest that even when initial extraction performs reasonably well, the bulk of human effort remains in post-extraction processing: cleaning inconsistencies, correcting misread values, aligning rows and columns, and validating outputs against source documents. The main bottleneck is not extraction itself, but the downstream effort to make the data

usable.

A key objective going forward is determining how much of this post-processing can be automated. Machine learning tools have strong potential here: natural-language errors such as mislabeled headers (producing high Levenshtein distances) could be identified and corrected using domain-tuned LLMs, and more sophisticated validation software could flag anomalies such as outliers or structural irregularities automatically – either correcting errors directly or generating targeted reports for human review, reducing manual workload while improving consistency and scalability.

6.4 Concluding Remarks

Taken together, this evaluation indicates that Textract, combined with the dataset-specific post-processing workflow described here, reproduced the manual reference transcription closely enough at the cell level, and preserved the downstream epidemiological quantities closely enough, to be a plausible option for this archive. It does not establish that the same workflow – or Textract generally – would perform comparably on other historical tabular sources. Researchers considering a similar workflow should treat these figures as a case study rather than a benchmark, and independently audit both extraction accuracy and downstream statistical quantities on their own material before relying on an automated extraction pipeline.

Acknowledgments

We would like to express sincere gratitude to Dr. Ben Bolker for his valuable direction and Noah Juravsky for the many insightful discussions that significantly accelerated the progress of this work.

References

1. Zhong X, ShafieiBavani E, and Jimeno Yepes A. Image-based table recognition: data, model, and evaluation. *Computer Vision – ECCV 2020*. Springer, 2020 :564–80. DOI: 10.1007/978-3-030-58589-1_34. Available from: https://doi.org/10.1007/978-3-030-58589-1_34
2. Hu J, Kashi RS, Lopresti D, and Wilfong GT. Evaluating the performance of table processing algorithms. *International Journal on Document Analysis and Recognition* 2002; 4:140–53. DOI: 10.1007/s100320200074. Available from: <https://doi.org/10.1007/s100320200074>
3. Mori S, Suen CY, and Yamamoto K. Historical review of OCR research and development. *Proceedings of the IEEE* 1992; 80:1029–58. DOI: 10.1109/5.156468. Available from: <https://doi.org/10.1109/5.156468>
4. Smith R. An Overview of the Tesseract OCR Engine. *Proceedings of the Ninth International Conference on Document Analysis and Recognition (ICDAR 2007)*. No DOI found. Washington, DC, USA: IEEE Computer Society, 2007 :629–33. Available from: <https://storage.googleapis.com/pub-tools-public-publication-data/pdf/33418.pdf>
5. Xia S, Xiong J, Dong H, Zhao J, Tian Y, Zhou M, He Y, Han S, and Zhang D. Vision language models for spreadsheet understanding: Challenges and opportunities. *arXiv preprint* 2024. DOI: 10.48550/arXiv.2405.16234. arXiv: 2405.16234 [cs.CV]. Available from: <https://doi.org/10.48550/arXiv.2405.16234>
6. Sahasrabudhe C, Ho Y, Winovich N, and Rajamanickam S. Imperfect Recognition: A Study of OCR Limitations in the Context of Scientific Documents. *2025 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*. 2025

:996–1002. DOI: 10.1109/IPDPSW66978.2025.00155. Available from: <https://doi.org/10.1109/IPDPSW66978.2025.00155>

7. Panhuis WG van, Grefenstette J, Jung SY, Chok NS, Cross A, Eng H, Lee BY, Zadorozhny V, Brown S, Cummings D, and Burke DS. Contagious diseases in the United States from 1888 to the present. *New England Journal of Medicine* 2013; 369:2152–8. DOI: 10.1056/NEJMms1215400. Available from: <https://doi.org/10.1056/NEJMms1215400>
8. Panhuis WG van, Cross A, and Burke DS. Project Tycho 2.0: a repository to improve the integration and reuse of data for global population health. *Journal of the American Medical Informatics Association* 2018; 25:1608–17. DOI: 10.1093/jamia/ocy123. Available from: <https://doi.org/10.1093/jamia/ocy123>
9. Sockett PN, Garnett MJ, and Scott C. Communicable disease surveillance: Notification of infectious diseases in Canada. *Canadian Journal of Infectious Diseases* 1996; 7:293–5. DOI: 10.1155/1996/279482. Available from: <https://doi.org/10.1155/1996/279482>
10. Totten S, Medaglia A, and McDermott S. Updates to Canadian Notifiable Disease Surveillance System. *Canada Communicable Disease Report* 2019; 45:257–61. DOI: 10.14745/ccdr.v45i10a02. Available from: <https://doi.org/10.14745/ccdr.v45i10a02>
11. Earn DJD, MacKinnon G, Manzin S, Roswell M, Cygu S, Shi C, Bolker BM, Dushoff J, and Walker SC. A century of weekly notifiable disease incidence data by province in Canada. *PLOS Global Public Health* 2025 Dec; 5:1–17. DOI: 10.1371/journal.pgph.0005550. Available from: <https://doi.org/10.1371/journal.pgph.0005550>
12. Stadler M, Bannert M, and Sailer M. Cognitive ease at a cost: LLMs reduce mental effort but compromise depth in student scientific inquiry. *Computers in Human Behavior*

- 2024; 160:108386. DOI: 10.1016/j.chb.2024.108386. Available from: <https://doi.org/10.1016/j.chb.2024.108386>
13. Amazon Web Services. OCR software, data extraction tool — Amazon Textract. 2025. Available from: <https://aws.amazon.com/textract/>
 14. Levenshtein VI. Binary codes capable of correcting deletions, insertions, and reversals. Soviet Physics Doklady 1965; 10:707–10. Available from: <https://api.semanticscholar.org/CorpusID:60827152>
 15. Wagner RA and Fischer MJ. The String-to-String Correction Problem. Journal of the ACM 1974; 21:168–73. DOI: 10.1145/321796.321811
 16. Wallinga J and Lipsitch M. How generation intervals shape the relationship between growth rates and reproductive numbers. Proceedings of the Royal Society B: Biological Sciences 2007; 274:599–604. DOI: 10.1098/rspb.2006.3754. Available from: <https://doi.org/10.1098/rspb.2006.3754>
 17. Ma J, Dushoff J, Bolker BM, and Earn DJD. Estimating initial epidemic growth rates. Bulletin of Mathematical Biology 2014; 76:245–60. DOI: 10.1007/s11538-013-9918-2. Available from: <https://doi.org/10.1007/s11538-013-9918-2>
 18. Jagan M and Bolker BM. epigrowthfit: Nonlinear Mixed Effects Models of Epidemic Growth. R package version 0.15.4. 2025. DOI: 10.32614/CRAN.package.epigrowthfit. Available from: <https://CRAN.R-project.org/package=epigrowthfit>
 19. Amazon Web Services. Amazon Textract Developer Guide. Amazon Web Services Documentation. 2025. Available from: <https://docs.aws.amazon.com/textract/latest/dg/what-is.html> [Accessed on: 2025 Sep 5]
 20. OfekNalian E. Investigating Table Extraction Tools for Historical Disease Data. James Stewart Research Award Report 2026. Available from: <https://mcmasteru365.sharepoint.com/sites/sci-public-docs/WEBMS/Forms/AllItems.aspx?id=%2Fsites%2Fsci%2Dpublic%2Ddocs%2FWEBMS%2FUndergraduate%2FStewart%5FFund%2F>

794 5FAward%5FWinners%2FEden%5FStewartAward%5FReport%5F2025%2Epdf&parent=
795 %2Fsites%2Fsci%2Dpublic%2Ddocs%2FWEBMS%2FUndergraduate%2FStewart%5FFund%
796 5FAward%5FWinners&p=true&ga=1

Supplementary Material

S1 Text. AWS Textract extraction script and setup instructions.

To run `AWS_Python_Table_Extraction.py`, first create a free AWS account at aws.amazon.com and sign in to the AWS Management Console. Go to the IAM service, create a new user (do not use root credentials), and attach a scoped inline policy granting only the action the script calls – `textract:AnalyzeDocument` – rather than the broad `AmazonTextractFullAccess` managed policy:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "textract:AnalyzeDocument",
      "Resource": "*"
    }
  ]
}
```

This scope was verified against the script's only Textract call, `client.analyze_document(...)`; if you extend the script to call other Textract operations (e.g. asynchronous jobs or `DetectDocumentText`), add the corresponding actions.

Generate an Access Key ID and Secret Access Key under that user's security credentials – save these immediately, as the secret key is shown only once. Install the AWS CLI and run `aws configure --profile textract-project` to store these credentials under a named profile rather than the default profile, entering the access key, secret key, and a default region (e.g., `us-east-1`, which matches the script's default). Set `AWS_PROFILE=textract-project` (or pass `profile_name` to `boto3.Session`) so the script authenticates as this scoped user; where your institution supports it, short-lived credentials issued via AWS IAM Identity

Center are preferable to long-lived access keys. Never commit credentials to version control, and rotate or delete the access key when the project ends. A free tier exists for Textract usage, with terms that vary by API operation and change over time; consult the current AWS Textract pricing page for the applicable limits and per-page rates before running a large extraction job.

Once credentials are configured, place your PDF or image files (PDF, PNG, JPG, or TIFF) in a single input folder and run the script from the command line:

```
python AWS_Python_Table_Extraction.py <input_dir> <output_dir>
```

This will create two CSV files per document – one with extracted table cell text and one with per-cell confidence scores – in the specified output folder. Because this script uses Textract’s synchronous AnalyzeDocument API, each input file is limited to a single page; multi-page PDFs should be split beforehand (e.g., with a tool like pypdf) or processed via Textract’s asynchronous API if full multi-page support is needed. See also [19].

S2 Text. Repository.

For repository containing scripts and figures used see our repository.

S3 Text. Textract JSON block structure.

At this point in the API call (section 2), Textract does not yet know “rows” or “columns”; it only sees shapes and text. Textract returns a flat list of BLOCK objects (not a table as recognized by humans). The response is one JavaScript Object Notation (JSON) array called **Blocks**. Each block contains, among other information, **BlockType**, which contains values such as **TABLE**, **CELL**, **WORD**, and **SELECTION_ELEMENT**. Each block also contains a confidence score.

A **BlockType** of “TABLE” represents one detected table and contains a list of cell ids. Each **CELL** block contains **RowIndex**, **ColumnIndex**, **RowSpan/ColumnSpan**, and relationships to **WORD** blocks, which contain the actual string value of the cell along with its own confidence value.

S4 Text. Levenshtein distance recurrence.

Let $s = s_1 s_2 \dots s_m$ and $t = t_1 t_2 \dots t_n$ be two strings of lengths m and n . The Levenshtein distance $D(i, j)$ between the characters $s_1 \dots s_i$ and $t_1 \dots t_j$ is defined by the standard dynamic-programming recurrence [15]:

$$D(0, j) = j \quad \text{for } j = 0, 1, \dots, n$$

$$D(i, 0) = i \quad \text{for } i = 0, 1, \dots, m$$

$$D(i, j) = \min \begin{cases} D(i-1, j) + 1 & \text{(deletion)} \\ D(i, j-1) + 1 & \text{(insertion)} \\ D(i-1, j-1) + \mathbf{1}_{s_i \neq t_j} & \text{(substitution)} \end{cases}$$

$$\mathbf{1}_{s_i \neq t_j} = \begin{cases} 0, & s_i = t_j \\ 1, & s_i \neq t_j \end{cases}$$

The Levenshtein distance between strings s and t is then $\text{Lev}(s, t) = D(m, n)$.

S5 Text. Preliminary exploration of a conversational LLM for table extraction.

Before settling on Textract, we informally tried extracting a small number of CANDID tables by prompting a general-purpose conversational large language model (ChatGPT) to read the scanned table and return its contents [20]. This was an unstructured, non-benchmarked pilot on a handful of tables rather than a controlled comparison: we did not evaluate it against a reference transcription, vary prompts systematically, or record error rates. Qualitatively, the tool produced frequent transcription errors that were tedious to locate and correct by hand, and we abandoned this approach in favour of Textract, a service purpose-built for structured table extraction. We do not draw any conclusion here about the underlying capability of conversational LLMs for this task in general – only that this particular informal attempt did not meet our needs, motivating the more systematic evaluation of Textract

873 reported in the main text.

874 **S1 Table. Per-lag cross-correlation values, Measles vs. Chickenpox.** Sign con-
875 vention: Difference = Textract – Manual. Data version: processed, no manual correction
876 (table 1). Summarized in table 3.

Lag	Manual	Textract	Difference
-12	0.0217	0.0244	0.0027
-11	0.1057	0.0871	-0.0186
-10	0.1760	0.1571	-0.0190
-9	0.2476	0.2199	-0.0277
-8	0.3334	0.3213	-0.0121
-7	0.4180	0.3996	-0.0184
-6	0.5087	0.4891	-0.0196
-5	0.5767	0.5315	-0.0453
-4	0.6284	0.5877	-0.0407
-3	0.6745	0.6264	-0.0482
-2	0.7473	0.7009	-0.0464
-1	0.8092	0.7766	-0.0326
0	0.8584	0.8329	-0.0255
1	0.8377	0.8078	-0.0298
2	0.8162	0.7682	-0.0480
3	0.7908	0.7385	-0.0524
4	0.7661	0.7412	-0.0248
5	0.7225	0.6668	-0.0557
6	0.6646	0.6102	-0.0544
7	0.6236	0.5536	-0.0699
8	0.5595	0.5395	-0.0199
9	0.4943	0.4662	-0.0281
10	0.4370	0.3991	-0.0379
11	0.3893	0.3386	-0.0507
12	0.3276	0.3091	-0.0185

877 **S2 Table. Per-lag cross-correlation values, Measles vs. Meningitis.** Sign conven-
878 tion: Difference = Textract – Manual. Data version: manually corrected (section 5.3).
879 Summarized in table 3.

880 **S3 Table. Per-lag cross-correlation values, Chickenpox vs. Mumps.** Sign conven-
881 tion: Difference = Textract – Manual. Data version: manually corrected (section 5.3).

Lag	Manual	Textract	Difference
-12	0.0236	0.0279	0.0042
-11	0.0258	0.0357	0.0100
-10	0.0055	0.0193	0.0138
-9	0.0291	0.0484	0.0193
-8	0.0348	0.0320	-0.0028
-7	0.0476	0.0392	-0.0084
-6	0.0052	0.0140	0.0087
-5	-0.0277	-0.0043	0.0234
-4	-0.0093	0.0064	0.0157
-3	-0.0301	0.0152	0.0453
-2	0.0050	0.0748	0.0699
-1	0.0594	0.1153	0.0559
0	0.0610	0.1168	0.0558
1	0.0384	0.0818	0.0434
2	0.0777	0.1366	0.0589
3	0.0848	0.1317	0.0469
4	0.0361	0.1110	0.0750
5	-0.0027	0.0588	0.0615
6	0.0188	0.0666	0.0478
7	0.0136	0.0299	0.0163
8	-0.0299	0.0004	0.0302
9	-0.0339	0.0110	0.0449
10	-0.0392	0.0077	0.0469
11	-0.0755	-0.0569	0.0186
12	-0.1223	-0.1041	0.0182

882 Summarized in table 3.

Lag	Manual	Texttract	Difference
-12	0.2937	0.2919	-0.0018
-11	0.3655	0.3638	-0.0017
-10	0.4384	0.4373	-0.0011
-9	0.4985	0.4994	0.0009
-8	0.5532	0.5535	0.0003
-7	0.6133	0.6112	-0.0021
-6	0.6720	0.6710	-0.0009
-5	0.7107	0.7082	-0.0026
-4	0.7554	0.7543	-0.0011
-3	0.7909	0.7857	-0.0052
-2	0.8246	0.8146	-0.0100
-1	0.8706	0.8703	-0.0002
0	0.8983	0.8974	-0.0009
1	0.8494	0.8448	-0.0046
2	0.7978	0.7878	-0.0100
3	0.7567	0.7448	-0.0119
4	0.7245	0.7115	-0.0129
5	0.6879	0.6734	-0.0145
6	0.6298	0.6177	-0.0120
7	0.5611	0.5511	-0.0100
8	0.4721	0.4643	-0.0078
9	0.3954	0.3896	-0.0058
10	0.3362	0.3265	-0.0097
11	0.2591	0.2457	-0.0135
12	0.1819	0.1677	-0.0143

Author Contributions

Conceptualization: Eden Ofek Nalian, David J.D. Earn, Steven C. Walker. Data curation: Eden Ofek Nalian. Formal analysis: Eden Ofek Nalian. Investigation: Eden Ofek Nalian. Methodology: Eden Ofek Nalian, David J.D. Earn, Steven C. Walker. Project administration: David J.D. Earn, Steven C. Walker. Software: Eden Ofek Nalian. Supervision: David J.D. Earn, Steven C. Walker. Visualization: Eden Ofek Nalian. Writing – original draft: Eden Ofek Nalian. Writing – review & editing: Eden Ofek Nalian, David J.D. Earn, Steven C. Walker.

Ethics Statement

This study used aggregate historical public-health records published by the Dominion Bureau of Statistics, comprising weekly counts of notifiable disease cases by province and territory. The data contain no individual-level or identifiable information, and no human participants were involved. Institutional review board approval was therefore not required.

Data Availability

The Canadian Disease Incidence Dataset (CANDID) used in this study is publicly available; see [11] for access instructions. All code (extraction, post-processing, and analysis scripts) and intermediate/derived data files underlying every figure and table reported in this manuscript are available in the public repository at https://github.com/EdenNal/Textextract_Results (also referenced as Supplementary Material, S2 Text). Raw Amazon Textextract API responses (JSON) and the exact AWS region/account configuration and extraction date(s) used to produce them are not preserved in this repository; only the processed CSV outputs of the extraction step are available (see Limitations, section 6.2).

Funding

Funding information for this manuscript is provided in the Financial Disclosure field of the journal’s submission system, per PLOS policy, rather than in the manuscript file.

Competing Interests

The authors have declared that no competing interests exist.